Data Science with Python Internship – Task 6

Advanced Machine Learning, Model Optimization & Feature Engineering

Objective

After completing:

- Task-4 → Data Visualization & Insights
- Task-5 → ML Pipeline + Logistic Regression Data Science with Python Int...

You will now move toward **advanced modeling and optimization**, which is essential in real-world data science.

The objective of this task is to:

- Improve model performance using **advanced algorithms**
- Apply **feature engineering techniques**
- Perform **hyperparameter tuning**
- Compare multiple models
- Build a **stronger and optimized ML pipeline**

Why This Task Matters (Industry Context)

- Real-world ML projects require:
  - More than one model
  - Optimization for best performance
  - Feature engineering for better accuracy
- Companies expect:
  - Model comparison
  - Performance tuning
  - Data-driven decision making

This task upgrades you from **baseline model → optimized ML system**.

What You Will Build

You will enhance your Titanic ML project with:

1. Feature engineering
2. Multiple ML models
3. Hyperparameter tuning
4. Model comparison
5. Performance optimization

Core Concepts You Will Learn

Concept	Explanation
Feature Engineering	Creating better input features
Model Comparison	Evaluating multiple models
Hyperparameter Tuning	Optimizing model performance
Overfitting Control	Improving generalization
Advanced Models	Random Forest, Gradient Boosting

Tools & Technologies

- Python
- Jupyter Notebook / Colab
- pandas, NumPy
- scikit-learn
- matplotlib / seaborn

Prerequisites

- Completion of Task-5
- Understanding of ML pipeline
- Basic evaluation metrics

Part A: Feature Engineering

Create new features from existing data.

Examples:

Python

```python
df["FamilySize"] = df["SibSp"] + df["Parch"]
df["IsAlone"] = (df["FamilySize"] == 0).astype(int)
```

Optional:

- Extract title from Name
- Age groups

Part B: Train Multiple Models

Train:

- Logistic Regression (baseline)
- Random Forest
- Gradient Boosting

Python

```python
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier

rf = RandomForestClassifier()
gb = GradientBoostingClassifier()

rf.fit(X_train, y_train)
gb.fit(X_train, y_train)
```

Part C: Hyperparameter Tuning

Use GridSearchCV:

```python
from sklearn.model_selection import GridSearchCV

param_grid = {
    "n_estimators": [50, 100],
    "max_depth": [3, 5, 7]
}

grid = GridSearchCV(RandomForestClassifier(), param_grid, cv=5)
grid.fit(X_train, y_train)

print(grid.best_params_)
```

Part D: Model Comparison

Compare:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Part E: Cross Validation

Use:

```python
from sklearn.model_selection import cross_val_score

scores = cross_val_score(rf, X, y, cv=5)
print(scores.mean())
```

Part F: Feature Importance

```python
import matplotlib.pyplot as plt

importance = rf.feature_importances_
plt.barh(X.columns, importance)
plt.show()
```

Part G: Final Model Selection

Choose best model based on:

- Performance
- Stability
- Interpretability

Expected Output

You should have:

- Improved ML model
- Better accuracy than Task-5
- Feature importance insights
- Optimized hyperparameters

Deliverables

Mandatory:

1. Jupyter Notebook
2. Feature engineering implementation
3. Model comparison results
4. Hyperparameter tuning output
5. Updated report including:
   - Improvements over baseline
   - Best model justification

Optional (Advanced)

- Use XGBoost
- Handle class imbalance
- Use pipelines with tuning
- Add model explainability

Free Learning Resources & Support Guide

(Feature Engineering, Model Comparison, Hyperparameter Tuning, Advanced ML)

Data Science with Python – Task 6

Free Learning Resources & Support Guide

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How to Use This Guide

To successfully complete Task-6, follow this approach:

- First understand **feature engineering concepts**
- Then learn **advanced ML models**
- Then apply **hyperparameter tuning**
- Finally perform **model comparison and optimization**

Use:

- YouTube tutorials
- Search-based learning
- Hands-on coding in Jupyter/Colab

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1. Feature Engineering (Start Here)

This is the most important step in improving model performance.

Search topics:

- Feature engineering in machine learning
- Creating new features in dataset
- Handling categorical and numerical features

Recommended YouTube Learning

https://www.youtube.com/watch?v=9rZq6yXx7eA

https://www.youtube.com/watch?v=3C6y4p3fQzE

These videos explain:

- Why feature engineering improves accuracy
- Real examples (like Titanic dataset)

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2. Random Forest (Advanced Model)

Search topics:

- Random Forest algorithm explained
- Decision trees vs Random Forest
- Random Forest sklearn tutorial

Recommended YouTube Learning

https://www.youtube.com/watch?v=J4Wdy0Wc_xQ

https://www.youtube.com/watch?v=ok2s1vV9XW0

These cover:

- Ensemble learning
- Feature importance
- Practical implementation

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3. Gradient Boosting (Advanced Model)

Search topics:

- Gradient Boosting explained
- Boosting vs bagging
- Gradient boosting sklearn

Recommended YouTube Learning

https://www.youtube.com/watch?v=3CC4N4z3GJc

https://www.youtube.com/watch?v=2xudPOBz-vs

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4. Hyperparameter Tuning (Very Important)

Search topics:

- GridSearchCV explained
- Hyperparameter tuning sklearn
- Model optimization techniques

Recommended YouTube Learning

https://www.youtube.com/watch?v=HdlDYng8g9s

https://www.youtube.com/watch?v=9GzGvX1pYhE

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5. Cross Validation (Reliable Evaluation)

Search topics:

- K-fold cross validation explained
- Cross validation sklearn
- Avoid overfitting ML

Recommended YouTube Learning

https://www.youtube.com/watch?v=fSytzGwwBVw

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6. Model Comparison & Selection

Search topics:

- Compare machine learning models
- Bias vs variance tradeoff
- Model evaluation techniques

Recommended YouTube Learning

https://www.youtube.com/watch?v=EuBBz3bI-aA

7. Feature Importance & Interpretation

Search topics:

- Feature importance Random Forest
- Model interpretability ML
- Explain ML models

Recommended YouTube Learning

https://www.youtube.com/watch?v=wpNl-JwwplA

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8. Overfitting & Regularization

Search topics:

- Overfitting in machine learning
- Regularization explained
- Bias vs variance

Recommended YouTube Learning

https://www.youtube.com/watch?v=6g4O5UOH304

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9. End-to-End Advanced ML Workflow

Search topics:

- Machine learning workflow step by step
- Titanic ML advanced project
- End-to-end ML pipeline

Recommended YouTube Learning

https://www.youtube.com/watch?v=i_LwzRVP7bg

https://www.youtube.com/watch?v=ukzFI9rgwfU

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10. Tools Setup & Practice

Learn:

- Google Colab usage
- Jupyter Notebook
- pandas, NumPy, sklearn

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11. Advanced Topics (Optional but Important)

Search topics:

- XGBoost explained
- LightGBM basics
- Model explainability SHAP
- Feature selection techniques

12. Practice Platforms

Use:

- Kaggle (Titanic dataset)
- scikit-learn datasets
- Open datasets

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13. Community Support

Search topics:

- Stack Overflow sklearn issues
- Reddit data science
- Kaggle discussion forums

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14. Recommended Learning Path (5 Days Plan)

Day 1

Feature engineering

Day 2

Random Forest + Gradient Boosting

Day 3

Hyperparameter tuning

Day 4

Model comparison

Day 5

Final optimization + reporting

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15. Essentials Checklist

Before completing Task-6, ensure:

- You can create new features
- You can train multiple models
- You can tune hyperparameters
- You can compare models properly
- You can interpret feature importance
- You can select best model logically